

Input Layer



Transaction Graph

$$G = (V, E, X)$$

$$\mathbf{X} \in \mathbb{R}^{|V| \times d}, \mathbf{A} \in \mathbb{R}^{|V| \times |V|}$$

Node features & adjacency



Network State

$$\mathbf{N}(t) = [\lambda, \beta, \tau]^T$$

λ : latency, β : bandwidth, τ : compute

Dynamic edge conditions

Network-Aware Attention Modulation



Attention Weights

$$e_{ij} = a(\mathbf{W}h_i, \mathbf{W}h_j)$$

$$\tilde{e}_{ij} = e_{ij} \cdot \gamma(\mathbf{N}(t))$$

γ : network-aware scaling function



Modulation Function

$$\gamma(\mathbf{N}) = \sigma(\mathbf{W}_\gamma \mathbf{N} + b_\gamma)$$

$$\alpha_{ij} = \text{softmax}_j(\tilde{e}_{ij})$$

σ : sigmoid, adaptive to network

Multi-Head Graph Attention (K Heads)



Head 1



Head 2

...



Head K



concat

GAT Output

$$h'_i = \frac{1}{K} \sum_{k=1}^K \sigma(\sum_{j \in N(i)} \alpha_{ij}^k \mathbf{W}^k h_j)$$

$K=8$ heads

$d=64$ dim

$L=3$ layers

Multigraph Discrepancy Module



Edge2Seq Encoder

$$e_{ij} = \text{MLP}([h_i \| h_j \| x_{ij}])$$

$$s = \text{BiLSTM}(\{e_{ij}\})$$

Hidden=256, Bi-directional



Discrepancy Computation

$$D(G_1, G_2) = \|s_1 - s_2\|_2$$

$$z_v = [h'_v \| s_v \| D_v]$$

Cross-graph structural difference

Trust-Weighted Output Layer



Trust Score

$$T_i = \alpha T_i^{\text{prev}} + (1-\alpha) \cos(\theta_i, \theta_g)$$

$\alpha=0.9$ (momentum)



Node Embeddings

$$\mathbf{Z} \in \mathbb{R}^{|V| \times d'}$$

$d' = K \times d$ (concatenated dim)

Classification



$$\hat{y}_v = \text{softmax}(\mathbf{W}_c z_v + b_c)$$

● Illicit ● Phishing ● Wash ● Legit

Notation: h : node embedding, α : attention weight, γ : network modulation, T : trust score, D : discrepancy